RAJNANDINI SINGH SOLANKI

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SUMMARY

AI & ML undergraduate (B.Tech, VIT Chennai) building LLM-powered, machine learning and full-stack systems for real-world problems. Smart India Hackathon finalist two years running (SIH 2024 Top 3; SIH 2025 Top 20 of 864+ teams), Microsoft Certified Cybersecurity Architect (SC-100), and researcher on LLM-based DDoS detection. Experience across data analytics, applied AI and computer vision; former Design Lead, GDG on Campus Game Development.

EDUCATION

B.Tech - Artificial Intelligence & Machine Learning · Vellore Institute of Technology, Chennai *Aug 2023 – Jul 2027*

Relevant Coursework: DSA · OS · Computer Networks · DBMS · OOPS · AI · ML · Computer Vision · Deep Learning · NLP · Cryptography & Network Security · Compiler Design · 3D Modelling · Web Programming

High School Diploma (PCM) · India International School Jaipur *2022*

EXPERIENCE

Research Intern · Savana *Oct 2026 – Nov 2026*

Data Analyst Intern · Bluestock™ *May 2026 – Aug 2026*

- Performed data analysis and generated insights to support business decisions at a fintech-focused platform.

AI Intern Trainee · Innaya Nutri Foods Private Limited *May 2026 – Aug 2026*

- Applied AI/ML techniques to food-tech workflows, supporting model development and data processing pipelines.

Design Lead, Gamedev Department · GDG on Campus, VIT Chennai *Jun 2025 – Jun 2026*

- Led end-to-end execution of Game-A-Thon 2025 - scheduling, coordination, and real-time leaderboard management for 200+ attendees.
- Designed and shipped two original games, boosting game-dev member participation by 40% and club outreach by 30%.
- Coordinated 30+ volunteers across event and creative tracks; served as host and emcee.
- Previously Game Development Member (Oct 2024 – Jun 2025); volunteered on the management committee for DevsHouse 2025, GDG VIT Chennai's flagship hackathon.

Development Member · Microsoft Innovations Club, VIT Chennai *Jun 2025 – Jun 2026*

AI Intern · E-Cell, IIT Madras *Nov 2023 – Jan 2024*

- Built an Object Detection model and a Vision Transformer for image classification using Python and deep learning frameworks.
- Applied ML concepts to real-world datasets; collaborated on computer vision workflows and model experimentation.

Office Bearer / Member · NSS VIT Chennai *Aug 2024 – Present*

- Increased volunteer participation by 40% through outreach and event initiatives.
- Led service activities during 7-day NSS Special Camp: student competitions, supply donations, fitness sessions, anti-plastic rallies, beach clean-ups.

President, Science Club · Student Council · India International School Jaipur *Apr 2021 – Mar 2023*

PROJECTS & HACKATHONS

Nyaya - AI Legal Research Engine *| Python · NLP · LLMs · KNN Similarity Search · TypeScript*

- Placed **Top 3 (SIH 2024)** out of 864+ teams. Built an AI-powered engine to aggregate legal data, extract key principles, and surface relevant precedents for commercial court dispute resolution.
- Paired a KNN similarity-search retrieval layer with a TypeScript web frontend for precedent lookup.

EduKnit - AI Achievement & Reporting Platform *| Python · AI · Full-Stack*

- Selected as one of **20 software finalists from 864+ teams** for SIH 2025. Platform unifies and verifies student achievements and automates institutional accreditation reporting.

RockSafe AI - Rockfall Prediction & Emergency Alert System *| Python · FastAPI · scikit-learn · React · TypeScript · Twilio*

- Built an ML risk-prediction service (Random Forest + SVM voting classifier) over 30+ geological, environmental and sensor features, served via a FastAPI backend.
- Integrated live weather and elevation APIs for site monitoring and tiered Twilio SMS evacuation alerts (Critical >75%, High 50-75%); React dashboard with animated risk visualization.

Financial Performance Metrics Toolkit *| Python · Jupyter*

- Python library for portfolio analytics: CAGR, Sharpe and Sortino ratios, alpha/beta, maximum drawdown and tracking error.

Research Paper Chatbot *| Next.js · TypeScript · OpenAI API · PDF Parsing*

- AI research assistant that ingests PDFs and enables conversational querying using streaming LLM responses, pdf-parse, jsdom, and OpenAI SDK server-side actions.

Pet Adoption System *| Node.js · Express · MongoDB · REST API*

- Full-stack adoption platform with pet registration, adopter management, Mongoose ORM, and tested REST API endpoints.

AES Steganography Tool | *HTML · JavaScript · Cryptography*

- Web-based tool combining AES encryption and image steganography to securely embed hidden messages within image files.

RESEARCH

From Text to Traffic: Repurposing Frozen Generative Models for DDoS Detection (Ongoing) | *Python · Transformers · Network Security*

- Investigating zero-shot DDoS detection across unseen network protocols using frozen LLMs; proposed architecture combines temporal flow tokenization, localized sequence normalization, and attention-based pooling.
- Exploring scale-invariant reasoning for cross-protocol cybersecurity analysis.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, PL/SQL, HTML/CSS
Frameworks & Tools: React, Next.js, Node.js, Express, FastAPI, Tailwind CSS, MongoDB, Git, Figma, Blender
AI / ML: Deep Learning, NLP, Computer Vision, Transformers, LLMs, PyTorch, scikit-learn, Predictive Analytics, OpenAI API
Domains: Full-Stack Development, Data Analysis, Game Development, Cryptography, AI Research

CERTIFICATIONS & COURSES

Microsoft Certified: Cybersecurity Architect Expert (SC-100) - Score: 882/1000 · Jul 2026
Microsoft Certified: Azure AI Fundamentals (AI-900) - Score: 828/1000 · Jan 2024
Entrepreneurship Program - Coincent | Innovation, leadership, and business fundamentals
Python 3 (Spoken Tutorial, IIT Bombay) - Score: 85% · Apr 2024

LANGUAGES

English (Advanced) · Hindi (Advanced) · Japanese (Intermediate) · French (Beginner)